

Anshul Singh

✉ anshul@uni-bonn.de | ☎ (+49) 152 35798064 | 🌐 anshulsc | 📄 Google Scholar | 🌐 anshulsc.github.io

RESEARCH INTERESTS

I work on making multimodal systems safe and predictable: understanding *how* vision-language models fail, *why* they fail at the level of internal representations, and whether those representations can be steered toward safer behaviour. My current work spans activation-level steering for cross-cultural safety grounding, adversarial robustness of multimodal models, and evaluation that exposes failure modes standard benchmarks hide.

Areas: AI safety & alignment • mechanistic interpretability • generative models of activations • activation steering • adversarial robustness • vision-language models • evaluation & benchmarking • multilingual & multicultural LLMs

EDUCATION

Apr 2026 Present	University of Bonn <i>M.Sc. Computer Science</i> Focus: AI safety and robustness of learning systems.	Bonn, Germany
Sep 2021 Jun 2025	Panjab University <i>B.E. Information Technology</i> · Grade: 9.17/10.0 (German equiv. 1.41)	Chandigarh, India

PUBLICATIONS

underline = me

[2026]	M4-RAG: A Massive-Scale Multilingual, Multi-Cultural, Multimodal RAG Benchmark [PDF] D. Anugraha, P. Irawan, <u>A. Singh</u> , E. S. A. Lee, G. Winata <i>retrieval-augmented generation, multilingual, multimodal, evaluation</i>	[CVPR 2026]
[2026]	HybridNet: Efficient Multimodal Fake News Detection S. K. Tah, <u>A. Singh</u> , L. Gupta, P. Katari, et al. <i>multimodal fusion, misinformation, efficiency</i>	[PP-MisDet @ CVPR 2026]
[2026]	ALARM: Multimodal Fake News Detection using MLLMs through Active Learning S. K. Tah, L. Gupta, <u>A. Singh</u> , et al. <i>active learning, MLLM, label efficiency</i>	[Under Review 2026]
[2025]	MTabVQA: Evaluating Multi-Tabular Reasoning of Language Models in Visual Space [PDF] [Page] <u>A. Singh</u> , C. Biemann, J. Strich <i>VLM, multi-hop reasoning, benchmark, GRPO post-training, synthetic data</i>	[EMNLP 2025]
[2025]	Lost in Translation and Noise: A Deep Dive into the Failure Modes of VLMs on Real-World Tables [PDF] <u>A. Singh</u> , R. Chaudhary, G. Singh, A. Kumar <i>robustness, failure analysis, multilingual, VLM evaluation</i>	[EurIPS 2025 Workshop, Oral]
[2023]	Comparative Analysis of State-of-the-Art Attack Detection Models [PDF] P. Kumari, V. Mangat, <u>A. Singh</u> <i>intrusion detection, feature selection, graph neural networks</i>	[ICCCNT 2023]

RESEARCH EXPERIENCE

Jun 2026 Present	HCDS, University of Hamburg 🌐 <i>Research Assistant</i> · Advisor: Prof. Chris Biemann - Leading SafeSymbol: testing whether VLMs recognise hazard, warning and prohibition symbols across cultural contexts, attribute the correct cultural origin, and reason about scene-appropriate use. - Probing where cultural priors are encoded and applying activation-level steering to internal representations, making safety-symbol grounding controllable rather than only diagnosable. - Designing an evaluation suite that separates perception failures from reasoning failures, so interventions target the responsible layer.	Hamburg, Germany
---------------------	--	------------------

Aug 2025 Feb 2026	IACV Lab, IISc Bangalore  <i>Research Associate</i> · Advisor: Prof. Soma Biswas - Built a data-efficient active learning framework for multimodal fake news detection that combines entropy-based uncertainty with LLM-guided disagreement, cutting labelling requirements while matching fully-supervised accuracy. - Studied adversarial robustness of MLLMs and diffusion models via activation steering, characterising which layers carry attack-sensitive directions; work produced two papers (CVPR-W 2026, one under review).	Bangalore, India
Jan 2025 Jun 2025	LT Research Group, University of Hamburg  <i>Research Intern</i> · Advisor: Prof. Chris Biemann - Created MTabVQA, the first benchmark for multi-hop reasoning over multiple table <i>images</i>, exposing a large accuracy gap between frontier VLMs and text-table baselines. - Applied GRPO post-training over multi-image table inputs, improving multi-hop accuracy over the SFT baseline. - Built a synthetic QA generation pipeline producing MTabVQA-Instruct, released for VLM table-reasoning fine-tuning.	Hamburg, Germany
Jun 2024 Sep 2024	MITACS GRI Fellow <i>Visiting Research Intern (Dalhousie University)</i> · Advisor: Prof. Ghader Manafiazar Predicted calving time and weaning behaviour from animal vocalisations with MFCC spectrograms and hybrid ViT-ResNet models (83% accuracy), using Attention-Guided CAM to verify the model tracked acoustically meaningful cues rather than recording artefacts.	Nova Scotia, Canada
Jun 2023 Jul 2023	Virtual Labs, IIT Roorkee <i>Machine Learning Intern</i> · Advisor: Prof. R. S. Anand Built a fault-detection pipeline for induction motors from vibrational signals; the analysis tooling supported three lab publications.	Roorkee, India

OTHER PROJECTS

- Cross-lingual Embedding Alignment for Indic Languages** *Python, PyTorch, FastText, SciPy* 
 - Hindi-English word-level alignment on FastText embeddings trained from Wikipedia dumps, reaching Precision@1 of 0.3464 vs. 0.3513 for supervised pre-trained baselines; extended with adversarial training for unsupervised alignment in low-resource Indic languages.
- Metal-FL: Cross-Platform Federated Learning** *Python, Kafka, gRPC, Socket.IO, Protobuf, PyTorch* 
 - Decentralised federated learning over Kafka and gRPC, aggregating weight updates from heterogeneous devices.

TECHNICAL SKILLS

Languages	Python, C++, Java, JavaScript, CUDA, SQL, $\LaTeX$
ML / Training	PyTorch, vLLM, veRL, DeepSpeed, FSDP/DDP, TRL, Ray, DSPy, LangChain, MLflow
Compute & Infra	Multi-node GPU and Intel Gaudi (HPU) clusters, SLURM, Docker, Kubernetes, AWS, PySpark
Systems & Data	FastAPI, Streamlit, Node.js, MongoDB, MySQL, Pinecone, Git

AWARDS, SERVICE AND LEADERSHIP

MITACS Globalink Award	CAD \$9,000 competitive research fellowship, Dalhousie University, Canada (2024).
Oral Presentation	EurIPS 2025 Workshop on AI for Tabular Data (selected talk).
Academic Standing	Graduated B.E. with 9.17/10.0 CGPA, Panjab University; ranked 3rd overall in the cohort.
Teaching	Instructor, IEEE workshop on the foundations of machine learning (100+ students).
ML/AI Lead	Google Developer Student Club, Panjab University led AI/ML initiatives and workshops.
Co-founder & Director	Uniqus Edutech Solutions, incubated at the RUSA Innovation Cell, Panjab University.
IEEE Student Branch	Executive Core Member; organizer, IEEE Conference on Women in Engineering (2022).